

ABDULLAH AL OMAR GALIB

Quantum Computing Researcher & AI Enthusiast



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Experience

UAM Competitive Programming & Research Association (CPrA)

Co-Organizer & Associate

June 2025 - Present, Madrid, Spain (Hybrid/Remote)

- Co-led MadQFF'25 (Madrid Qiskit Fall Fest 2025), a 3-week international quantum computing festival at Universidad Autónoma de Madrid, attracting global participants through a hybrid format.
- Coordinated 20+ expert speakers from IBM Quantum, Pasqal, Multiverse Computing, and ICFO, and facilitated 7+ hands-on workshops on quantum algorithms, ML, and applications.
- Organized a 4-day hackathon with daily challenges, creating original problems to test participants' skills in building quantum applications with Qiskit.
- Contributed as web developer for the official event website, ensuring seamless registration and resource access.

Quantum High School Organization

Co-Founder & Vice President

July 2023 - August 2025, Global

- Spreading quantum education globally through online initiatives, including webinars, workshops, and interactive content.
- Mentored 500+ students in quantum computing, creating tailored lesson plans and interactive content.
- Organized and led two highly successful quantum education events. The first event was dedicated to teaching quantum computing to beginners, ranging from high school students to graduate students, and was attended by over 300 participants. The second event focused on cutting-edge quantum computing research. Both events received positive feedback and contributed to the growth of quantum education and research.

QWorld

Research Intern

July 2023 - August 2023, Remote

- Researched classical and quantum approaches for Quantum State Tomography (QST).
- Implemented and compared different Quantum Machine Learning (QML) approaches for QST, such as supervised and unsupervised learning techniques, to assess their performance and accuracy. Utilized cutting-edge quantum hardware and quantum simulators to conduct QST on multi-qubit networks and complex entangled states.
- Achieved an impressive 98% fidelity in the quantum state reconstruction process, validating the effectiveness and precision of the chosen methodologies.
- The research findings and results were documented in a comprehensive research paper, contributing to the scientific community's understanding of quantum state tomography.

Projects

- **Bronze Pytket (Quantinuum)** - An introductory quantum computing tutorial using Pytket, created in August 2023 with Quantinuum support.
- **Quantum Convolutional Neural Networks vs Fully Classical CNNs** - Compared hybrid quantum-classical CNNs with classical CNNs on MNIST, achieving 90.54% accuracy with the quantum model, outperforming the classical model by 8.57%.
- **Driving Safety (Machine Learning)** - Created a real-time driving safety application using facial landmark detection and traffic sign recognition, achieving 99% accuracy.
- **Quantum Computing Notebook for Beginners** - Authored a beginner-friendly Jupyter notebook using Qiskit to introduce fundamental quantum computing concepts.
- **VTOL Vision: Multi-Class Object Detection** - Developed a real-time 48-class object detection system using YOLOv8 for geometric shapes, digits, and letters, with a custom dataset and style-transfer pipeline, optimized for edge deployment on Raspberry Pi 4, achieving 40.6% mAP@50.
- **Fruit Detection and Classification** - Created an automated image recognition system using YOLOv8 to detect and classify 36 classes of fruits and vegetables with 91.68% mean Average Precision (mAP). Employing computer vision and machine learning techniques, it is optimized for real-time use in agricultural monitoring, quality control processes, inventory management, and educational tools to improve efficiency in produce handling.
- **Nighthawk: AI-Powered Modular Security Tool Orchestrator** - Built an AI-powered system to orchestrate modular security tools, facilitating integrated management of cybersecurity components for threat detection and response in dynamic environments.
- **Quantum Tunneling Simulation with QuTiP and Manim** - Developed a simulation to model and visualize quantum tunneling phenomena, utilizing QuTiP for solving quantum dynamics equations and Manim for creating animated representations of wave functions and particle behaviors.
- **C_Flex: A Collection of Whimsical Python Tools** - A creative coding playground featuring diverse Python utilities including a terminal-based Markdown previewer, AI tools, Conway's Game of Life, and media downloaders, utilizing Rich and Curses libraries for UI with cross-platform compatibility.

Open Source Contribution

- **Mitig** - Authored tutorial on error mitigation pipeline integrating Pauli Twirling, Digital Dynamical Decoupling, Readout Error Mitigation, and Zero-Noise Extrapolation using Cirq.
- **PauliPropagation.jl** - Developed Python usage example notebook for Julia-based Pauli propagation library for quantum error analysis.
- **Qualtran** - Added comprehensive docstrings to fundamental quantum gates (CZPowGate, Rx, Ry) in Google's quantum circuit library.
- **scqubits** - Enhanced documentation with detailed function explanations and updated example notebooks for superconducting qubits simulation.

Publications

N. Innan, O.I. Siddiqui, S. Arora, et al. Quantum state tomography using quantum machine learning. *Quantum Machine Intelligence*, 6(1):28, 2024. doi:10.1007/s42484-024-00162-3.

M. A. Khan, N. Innan, A. A. Galib, and M. Bennai. Brain tumor diagnosis using quantum convolutional neural networks. *ArXiv*, 2024. URL: <https://arxiv.org/abs/2401.15804>.

Skills

Quantum Computing: Qiskit, Pytket, Cirq; **AI & Machine Learning:** TensorFlow, Keras; **Software Development:** Unreal Engine 5, Python; **Community & Education:** Webinars, Workshops, Lesson Planning, Mentoring; **Project Management:** Project Leadership, Team Collaboration, Time Management; **Soft Skills:** Leadership, Communication, Collaboration, Problem-Solving

Education

BRAC University

Dhaka, Bangladesh

Bachelor of Science - Theoretical Physics

Currently pursuing

St. Joseph Higher Secondary

Dhaka, Bangladesh

School

Higher Secondary School Certificate (HSC)

Result: GPA 5.00

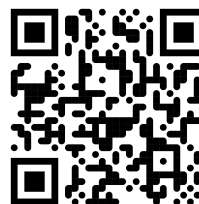
Languages

- Bangla (Native)
- English (Advanced)

Awards & Honors

- **Winner, IonQ Challenge – MIT iQuHack 2026 (Remote)**, Issued by MIT & IonQ · 2026
Won the remote portion of MIT's iQuHack 2026 IonQ Challenge by designing LOCC circuits for quantum entanglement distillation (using BBPSSW and DEJMPS protocols) on IonQ hardware. Awarded \$300 Amazon gift card.
- **Qiskit Advocate**, Issued by IBM Quantum · Aug 2025
- **Bounty Winner, unitaryHACK 2025**, Issued by Unitary Fund · May 28 - June 11, 2025
Successfully claimed \$525 in bounties by resolving 5 issues across 5 open-source quantum projects (Mitiq, PauliPropagation.jl, Qualtran, QuantumArtHack, scqubits), contributing to tutorials, documentation, and feature enhancements in the quantum software ecosystem.
- **2nd Runner-Up, YQuantum Hackathon**, Issued by Yale University & BlueQubit
Placed as 2nd runner-up in Yale University's premier quantum computing hackathon, competing in challenges focused on advanced quantum algorithms and applications.
- **Winner, Asia Pacific Regional Ozone2Climate Art Contest** · Sep 2022
Won regional art competition (Mahidol University, UNEP, UNESCO) promoting ozone protection and climate action; awarded 15,000 Taka.

Certificates



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